

01 — Departure from CSR: $\tilde{\delta}$ and null tables

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1 Theory

Setting

- Window $W = [0, 1]^2$
- $\bar{n}$ = expected number of points in W .
- Point cloud $x = \{x_1, \dots, x_n\} \subset W$; $n(x) = |x|$ denotes realised number of points

Estimator

- Centred L -function of a pattern x , built from Ripley's K function:

$$\hat{\phi}_x(r) = \sqrt{\hat{K}_x(r)/\pi} - r, \quad (= 0 \text{ under CSR})$$

- Evaluated on a fixed grid $0 < r_1 < \dots < r_K = 0.25$, giving $\hat{\phi}_x \in \mathbb{R}^K$

Radius set

Everything in this subsection is defined for fixed $n(x)$. The test and $\tilde{\delta}$ only use radii in

$$R(n) = \{r_k : \mu(r_k; n) \geq p\} = [r_{\min}(n), 0.25], \quad \mu(r; n) = \frac{1}{2}n(n-1)\pi r^2,$$

i.e. radii with at least p expected pairs closer than r under $\mathbb{P}_{0,n}$.

- This is because with few pairs, $\hat{\phi}$ takes a handful of values and the standardised deviation is heavy-tailed (at $n = 99$, $\mathbb{P}(|z| > 3)$ is $10\times$ Gaussian for $\mu < 0.3$, $3\times$ for $\mu \in [3, 10)$). Such radii win the null maximum too often, raising c_{95} and costing power at every other radius
- Binding constraint is the smoothness of $c_{95}(n)$, needed for the spline across n . Below $p \approx 5$ the raw c_{95} has structure at large n that no spline follows:

p	c_{95} mean z^2 (expect ≈ 1)	residual s.d.	Monte Carlo noise
10	0.77	0.019	0.022
5	0.97	0.023	0.022
4	1.62	0.036	0.029
3	5.22	0.040	0.026

- Use use $p = 5$. Then

$$r_{\min}(n) = \sqrt{\frac{2p}{\pi n(n-1)}} \approx \frac{1.78}{n}$$

Null distribution

We compute the null distribution at every $n(x)$ on a grid.

- Generate 12 000 CSR clouds from $\mathbb{P}_{0,n}$
- Estimate null moments for $r \in R(n)$ using half of the CSR clouds:

$$m_0(r; n) = \mathbb{E}_{0,n}[\hat{\phi}_X(r)], \quad s_0(r; n) = \text{sd}_{0,n}[\hat{\phi}_X(r)]$$

- Using remaining half of CSR clouds, compute maximum deviation of each cloud x :

$$M_n(x) = \max_{r \in R(n)} \frac{|\hat{\phi}_x(r) - m_0(r; n)|}{s_0(r; n)}$$

- Determine critical value: $c_{95}(n) = 95\%$ quantile of M_n over these clouds.
- Repeated for every $n(x)$ on grid, then smooth m_0 , s_0 , c_{95} across $n(x)$ (cubic splines in $\log n$) $\Rightarrow$ less Monte Carlo noise.

Studentised global envelope (Myllymäki et al. 2017)

- Reject CSR at 5% iff $M_n(x) > c_{95}(n)$, $n = |x|$; size 0.05 by definition of c_{95} (up to Monte Carlo error)
- We define the "distance from CSR" $\tilde{\delta}$:

$$\tilde{\delta}(\theta, \bar{n}) = \frac{1}{c_{95}(\bar{n})} \max_{r \in R(\bar{n})} \frac{|L_\theta(r) - r|}{s_0(r; \bar{n})}$$

- Same functional as M_n , applied to the closed-form L_θ at $\bar{n}$, uncentred ($m_0 = 0$)
- $\tilde{\delta}(\text{CSR}) = 0$ exactly and $\tilde{\delta} = 1$ is on the rejection boundary
- Units: null s.d.s, relative to the critical value
- This property is known before simulation, so is usable as a sweep coordinate
- At fixed shape, $\tilde{\delta}$ grows with $\bar{n}$ (s_0 shrinks)
- Depends on the cutoff when the departure lives below $r_{\min}$: a hard core of radius $R < r_{\min}$ enters $\tilde{\delta}$ only through the pair deficit it leaves above $r_{\min}$, diluted by the CSR count there, so its $\tilde{\delta}$ scales with p

2 Implemented pipeline

Code

src/cloudforger/classical/lfunction.py	$\hat{K}$, $\hat{L} - r$, weights, $L - r$ from $K - \pi r^2$
src/cloudforger/departure/config.py	config, n grid, paths
src/cloudforger/departure/simulate.py	null patterns $\rightarrow$ stored curves
src/cloudforger/departure/fit.py	curves $\rightarrow$ spline tables
src/cloudforger/departure/tables.py	tables $\rightarrow M_n, c_{95}, \tilde{\delta}$
src/cloudforger/departure/validate.py	size on fresh patterns
scripts/departure.py	CLI: <code>simulate</code> , <code>fit</code> , <code>validate</code>
configs/departure/	<code>config.yaml</code> , <code>tables.npz</code> , <code>report.json</code>

Configuration

- n grid: $\text{round}(20 \cdot 1.03^k)$ up to 2000 $\Rightarrow$ 153 values
- Per grid n : 6000 fit + 6000 calibration patterns
- Validation: 50 000 patterns, n log-uniform integer on $[20, 2000]$
- $p = 5$ pairs; $\alpha = 0.05$; knots: 8 interior (moments), 4 interior (c_{95})

Steps

1. **Simulate**: null curves for every grid n , stored
2. **Fit**: moments and c_{95} per n , splines in $\log n \rightarrow \text{tables.npz}$
3. **Validate**: size on fresh patterns at random n
4. **Relabel** (`scripts/relabel.py`): after a refit, recompute $\tilde{\delta}$ for every swept θ under the new tables; manifests keep `delta` (the sampling target) and `delta_tilde` (current, what regimes use)

Results

Check	Result	Reference
Fitted m_0 vs raw, mean z^2	0.94	≈ 0.92 (1 – params/grid)
Fitted s_0 vs raw, mean z^2	0.96	≈ 0.92
Max relative error of s_0	4.0%	—
Fitted c_{95} vs raw, mean z^2	0.97	≈ 0.95
c_{95} range; max s.e.	2.87–3.28; 0.030	—
Size, 50 000 patterns	0.0518 ($z = 1.8$)	0.05
Size by n band	0.049–0.056, all $ z \leq 2.4$	0.05
Size, 20 000 Poisson sweep patterns	0.0496	0.05
Centring gap max $ m_0 /s_0/c_{95}$	0.026–0.044	—

Estimator checks

- `tests/test_lfunction.py`:
 - Weights vs brute-force circle fraction: $|\Delta| < 10^{-3}$
 - Two points at distance 0.1: $\hat{K} = 0$ below, $= 1$ above
 - $\mathbb{E}\hat{K} = \pi r^2$ under binomial (400 patterns, within 3%)
 - One-off: agrees with the previous implementation (`baselines/vihrs.py`) to 10^{-9}
- Further statistical notes: `01_departure_notes.md`.